# Tech Bites

**A complete, plug-and-play Technician-license study program in 24 short modules -- built to recruit new hams by dropping into the meetings a group already holds.**

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## What This Is

Tech Bites is the entry-level member of the Bites family – the sibling of General Bites and Extra Bites, retargeted for the FCC **Technician** license, the first amateur radio license. It is **not** a class. It is a sequence of short concept lessons plus printable self-study sheets that moves a complete beginner to a passing grade on the 35-question Technician exam.

But Tech Bites has a different job from its two siblings. General Bites and Extra Bites serve people who are **already licensed hams**, and are pitched to radio clubs. Tech Bites is built for people who are **not hams yet** – it is a recruitment tool for the hobby itself.

Because a "Bite" is a meeting-friendly 5-10 minute segment, Tech Bites drops into the regular meetings of almost any existing group – a Scout troop, a 4-H club, a fire department, a school club – as a small add-on that does not disrupt their normal business. Over a single season of that group's meetings, its members can earn their Technician licenses together.

## The Two Layers

| Layer | Length | Job | Where it lives |
|---|---|---|---|
| **Tech Bite** | 5-10 min | One compact concept, taught from scratch with plain language, a diagram, and one example. Assumes no prior radio knowledge. May cover one exam group or a natural pair. | A short segment at any group meeting. Handout goes home. |
| **Tech Dessert** | self-study | Rules, privileges, operating procedures, safety – memorization-heavy material that delivers fine as a take-home reference. May cover one group or a natural pair. | Distributed alongside the Bites. Done at home. |

Like General Bites, Tech Bites is **two layers only** – 12 Bites and 12 Desserts. There are no longer "Course" sessions; Technician material is light enough to deliver well in short Bites.

## Pass Math For The Technician Exam

| Quantity | Working assumption |
|---|---|
| Pool size | 409 active questions across 35 exam groups |
| Exam length | 35 questions, one drawn per group |
| Pass threshold | 26 of 35 correct (74%) |
| Exam questions on covered material | 35 of 35 – the whole pool is covered |
| Accuracy on covered material | about 85% |
| Expected score | 35 × 0.85 ≈ **30** |

That projects to about 30 correct – roughly **four above** the 26-question pass line, a very comfortable margin. Because Technician rules and operating questions are plain recall, real-world cohorts typically exceed the 85% projection.

# Plan: 12 Bites + 12 Desserts

All 35 exam groups are covered. Four Bites and six Desserts each own a natural pair of related groups; one Bite (the math) owns three; the rest own one group each.

## 12 Tech Bites (one concept lesson per meeting)

| Slot | Title | Groups | Qs |
|---|---|---|---|
| B1 | How radio waves travel and behave | T3A + T3B | 24 |
| B2 | Propagation: skip, scatter, and the radio horizon | T3C | 11 |
| B3 | Setting up your station | T4A | 12 |
| B4 | Electricity basics: voltage, current, power | T5A | 11 |
| B5 | The math, made small | T5B-T5D | 39 |
| B6 | Components: resistors, capacitors, inductors, semiconductors | T6A + T6B | 23 |
| B7 | Reading schematics, and what circuits do | T6C + T6D | 23 |
| B8 | Inside the radio: the transceiver | T7A | 11 |
| B9 | Measuring SWR and the antenna system | T7C | 11 |
| B10 | Modes and bandwidth: FM, SSB, CW | T8A | 12 |
| B11 | Working amateur satellites | T8B | 12 |
| B12 | Antennas: gain, polarization, and types | T9A | 11 |
| **Total** | **12 Bites** | **17 groups** | **200** |

## 12 Tech Desserts (self-study handouts)

| ID | Title | Groups | Qs |
|---|---|---|---|
| D1 | The Amateur Service and your frequency privileges | T1A + T1B | 23 |
| D2 | Your license, and what you may transmit | T1C + T1D | 23 |
| D3 | Radio controls (companion to Bite 3) | T4B | 11 |
| D4 | Control operator, station ID, and repeaters | T1E + T1F | 22 |
| D5 | Making contacts and using repeaters | T2A + T2B | 25 |
| D6 | Public service and emergency operating | T2C | 12 |
| D7 | Electrical and tower safety | T0A + T0B | 23 |
| D8 | Fixing interference (companion to Bite 8) | T7B | 11 |
| D9 | Using a multimeter, and soldering (companion to Bite 9) | T7D | 11 |
| D10 | Digital and internet-linked operating | T8C + T8D | 23 |
| D11 | RF exposure safety | T0C | 13 |
| D12 | Feed lines and connectors (companion to Bite 12) | T9B | 12 |
| **Total** | **12 Desserts** | **18 groups** | **209** |

## System Totals

| Layer | Groups | Pool Qs |
|---|---|---|
| Bites | 17 (12 modules) | 200 |
| Desserts | 18 (12 modules) | 209 |
| **Pool covered** | **35 of 35** | **409 of 409** |

# Running the Program

## Modular Compression

The 12-meeting length is the **gentlest** setting, not a fixed duration. Every Bite and Dessert is an independent module, so a group dials the pace to fit – same materials, same handouts, same recordings, only the schedule changes.

| Pace | Bites / mtg | How it runs | Finishes in |
|---|---|---|---|
| **Standard** | 1 | One short Bite per weekly meeting. Relaxed; never crowds the group's own business. | 12 weeks |
| **Accelerated** | 2 | Two Bites per session – only 10-15 min of program time. | 6 meetings |
| **Study-Group Sprint** | 3, weekly | Three Bites per weekly session. Still well under an hour per session. | 4 weeks |

Most groups that Tech Bites is aimed at – Scout troops, 4-H clubs, school clubs – meet **weekly**. At one Bite per weekly meeting the program runs about three months; doubled up it is six weeks, and a focused study group finishes in four. For this system, the compressed paces are the normal mode, not the exception.

## Plug and Play

Tech Bites includes a **pre-recorded audio presentation for every Bite**. A volunteer facilitator distributes the handouts and presses play on a phone or a small speaker – no one has to teach, and no one has to be a ham. The recordings do the teaching; the Desserts go home. The whole system is free and self-contained.

## What The System Will Not Do

- It will not turn a group's meeting into a school – each Bite is a short, optional add-on.
- It will not replace practice questions. Concept exposure plus repeated practice-question drill is the working combination; free practice exams are at eham.net/exams.
- It will not teach beyond the Technician exam. It is the on-ramp; General Bites and Extra Bites continue the journey.

## Success Target

Move a complete beginner – a Scout, a 4-H member, a curious adult with no radio background – from zero knowledge to a confident pass on the 35-question Technician exam, over a single season of the meetings their group already holds. Done well, Tech Bites does not just produce a license; it produces a new ham, and a new member for the hobby.